

A Visual Approach for the Comparative Analysis of Character Networks in Narrative Texts

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ABSTRACT

The analysis of a novel's plot and characters are challenging and time-consuming tasks in literary criticism. Typically, humanities scholars want to describe and compare characters' personality traits, their roles, their relationships, and the evolution of these aspects over the course of a novel. Nowadays, due to the digitization of literature, humanities scholars can be supported in these endeavors with computational methods. In this paper, we present an approach that offers several means to analyze the plot and characters of a novel visually. Analysts can easily switch between an adjacency matrix and a node-link representation, which provide an overview of the characters and the relationships between them. Both views enable analysts to select different text ranges of the novel for studying the commonalities and differences of the character constellations within these ranges. We offer interactive visual representations to help investigate the relationships between the characters in more detail. Additionally, we link the visual representations with the novels' texts to support the inspection and verification of previously generated ideas and hypotheses. To demonstrate the benefits and limitations of our approach, we present two usage scenarios. The first one is based on a fictitious analysis and the second one discusses applications that were carried out during joint workshops with humanities scholars. Finally, we present and discuss the insights gained by an expert study and the design decisions of our approach.

Keywords: Visual text analysis, document analysis, close reading, distant reading, digital humanities, graph comparison.

1 INTRODUCTION

In recent years, digital humanities (DH) has become a much-discussed and exciting research field for computational linguistics, humanities scholars, and visualization researchers. A typical task of humanities scholars from the field of literary studies is character analysis in novels. For example, they want to characterize the main protagonists of a plot, to collect information about the relationships of the characters, or to compare different character constellations and plot events with each other [34]. Such tasks require manual annotations and complex analysis, which are very time-consuming. The digitization of literature and the digital libraries which provide easy access to those digital copies open up new possibilities to support these analysis tasks [24]. In this context, computational linguistics and visualization researchers come into focus, since the combination of natural language processing (NLP) and visualization techniques are of great appeal for humanities scholars.

With the aid of NLP methods, it is possible to extract named entities [39] or the attitude and evolution of the characters in a plot [45]. The extracted information can then be visualized as “distant reading”-abstractions of a text [32]. Such visual abstractions, for example, word clouds [25] or maps [14] provide information that can serve as a basis for new thoughts and hypotheses. However, it is evident that “close reading” will always be an essential analysis step for humanities scholars. Therefore, it is necessary to link the visual abstractions with the text in order to give the humanities scholars the possibility to examine and verify their assumptions.

We developed our approach in a formative process in the context of the DH project CRETA [1]. This project includes researchers from the disciplines of computational linguistics, visualization, and many different fields in the humanities. The project's goal is to develop general workflow methodologies and approaches to investigate named entities and their related concepts in texts. The participating humanities scholars use annotation systems to annotate these entities manually, whereas the computational linguists support the annotation workflow with automatic methods. Thus, we can develop our approach with gold-labeled annotations and automatically generated annotations for texts where no manual annotations are available. Automatic approaches offer a good way to jump-start analysis of a literary text and provide a first overview. However, they often do not meet the literary scholars' requirements regarding a thorough analysis for two reasons. Out-of-the-box NLP methods trained on newspaper or journal texts do not achieve good recognition rates on literary texts and even task-tailored recognition methods cannot deliver the perfect results expected by scholars for detail analysis. The text documents in this DH project are mostly novels¹ that comprise of several parts or different versions of the text. Examples are *The Sorrows of Young Werther* by J. W. von Goethe [42] or *Parzival* by W. von Eschenbach [41]. The humanities scholars are interested in investigating and comparing character constellations at different timestamps of the plot. The either manually or automatically derived characters networks of the narrative texts are typically smaller compared to social media networks. Therefore, the quality and evolution of the characters and their relationships are of interest.

Now, in order to support such an analysis, interactive visualization can be employed. We propose an interactive approach that focuses particularly on the comparative analysis of character networks. Users can freely select arbitrary ranges of a text document and visually compare the networks of the characters contained therein. Regarding the cooperation with our humanities collaborators, we decided to use established visualizations with which they are already familiar or that are easy to understand. Users can switch seamlessly between a node-link diagram for smaller and an adjacency matrix for larger character networks. Both visualizations provide a good overview of the characters and the relationships between them and have their benefits and shortcomings for different analysis tasks [4, 17]. Our approach provides interactive features and visual text abstractions that enable users to inspect the relationships of the characters in more detail. Additionally, we also link the

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¹In this work, we use the term *novel* interchangeably with *narrative texts* knowing that is a simplification.

visualizations with a text view to support the humanities scholars in their general workflow.

The main contributions of this work are the following:

- We contribute an interactive visualization approach that enables users to select different ranges of a text document and to visually analyze and compare the character networks occurring within them.
- The comparative approach provides interactive and visual abstractions which facilitate the analysis of qualitative and semantic information about the change of relationships between the characters as well as the analysis of the corresponding text passages.
- We discuss the insights of an domain expert study and the design decisions of our approach.

2 RELATED WORK

Our approach deals with the visualization and visual comparison of graphs as well as with visual text analysis and the representation of characters mentioned in texts. In the following, we discuss previous works from these fields and how they influenced our approach or differ from it, but restrict the detailed discussion to works that are closely related to ours either concerning the applied visual interactive techniques or regarding the extraction of characters from texts.

2.1 Visual Graph Comparison

Matrix-based and node-link visualizations are an established way to represent graph structures [6]. Both approaches were discussed in surveys, were evaluated regarding different aspects in controlled user studies, and were presented in many variations for a multitude of tasks.

For example, Alper et al. [4] conducted a user study based on the comparison between a matrix-based and node-link visualization. Additionally, they investigated the design space for the visual encoding of vertices and edges in both visualizations. Ghoniem et al. [17] examined the readability of a node-link diagram compared to a matrix-based visualization. For the experiment, they defined typical tasks for graph analysis and evaluated the results statistically.

Landesberger et al. [43] reported on automatic methods, visual representation, and interactive features that support the analysis and exploration of large graphs. Gleicher et al. [18] presented a survey that investigates the design space for the visual comparison. They found three main categories, which Beck et al. [6] slightly refined in their survey for dynamic graph visualization. The categories comprise “juxtaposed”, “superimposed”, and “integrated with explicit visual encoding”. Juxtaposition approaches depict different states next to each other, whereas the superimposed technique presents the different states on top of each other [18]. In an integrated approach, the selected sequences, text passages in our case, are an integral part of the visualization [6].

There exist many approaches that use superimposition. For example, Jianu et al. [23] compared experimental results and user-constructed models visually using such an approach. Groh et al. proposed the DySoN [19] system, which uses a superimposed node-link approach with stacked layers that represent the different states of the dynamic graph.

Another class of techniques uses juxtaposition to compare graphs. Albers et al. developed the Sequence Surveyor [3] system that provides an overview of genome alignment data. Here, the different color-coded sequences of genes are shown in a juxtaposed way. Blascheck et al. [7] place graphs side by side and use an explicit encoding to support the comparison of eye movement data.

Integrated approaches with explicit visual encoding are widely used as well. Reitz [36], for example, introduced a framework that

supports the analysis of time-related ego networks. It offers different visual encodings for the vertices and edges to represent information of different timestamps. Another integrated approach that focuses on ego networks was presented by Shi et al. [38]. The ego node represents a timeline and the outgoing position of each edge depicts when an edge first appears.

Finally, there exist also combinations of the three mentioned categories. For instance, Brandes et al. [9] combined juxtaposed and superimposed representations for the comparison of proteomic networks. Federico et al. [16] offered three different views (juxtaposed, superimposed, and two-and-a-half-dimensional) in one system to combine all the advantages of the different techniques. Concerning this categorization, our work also provides different visual representations that comprise the categories juxtaposition and integrated with explicit visual encoding.

Since we deal with dynamic networks, time obviously plays a role in the context of the presented approach. However, time in narrative texts has different aspects such as narrated time and narration time. Narration time means the time in which the storyline is read by the reader, whereas narrated time means the time course of the storyline that the author has created. In our approach, we consider only narration time, and thus we ignore temporal compression, relaxation, or even temporal rearrangement of the narration. Regarding the different ways of modeling time as Aigner et al. [2] describe them, we use an interval model.

2.2 Visual Character Network Analysis

In recent years, many visual analytics approaches have been developed that deal with social network analysis [10, 20, 46]. We focus on a specific form of social network analysis that deals with the visual analysis of character or figure networks derived either manually or automatically from narratives in film or text. These data sources are “unstructured” from the perspective of computer science. While sizes of character networks are typically smaller, the quality and evolution of the characters’ relations are of interest. Accordingly, we concentrate the discussion of related work specifically on the field of character network analysis [37].

The following approaches can be found in the focus of the field of research outlined above. El-Assady et al. introduced NEREx [15], an approach that supports the visual comparative analysis of discussions between several persons. It automatically extracts named entities and information about their relationship using NLP methods, and it offers a node-link visualization to explore them visually. Kim et al. [27] proposed an approach that uses word clouds in combination with node-link diagrams for visualizing entities and relationships in text corpora. However, the approach does not scale for larger networks and does not take into account the temporal information. Kurzhals et al. [29] presented a visual analytics approach that enables users to investigate extracted characters from movies on multiple levels of detail. A very well-known approach that supports the exploration of named entities is Jigsaw [39]. Here, several coordinated views enable users to trace named entities and explore their relationships through a large text document collection.

Furthermore, there are approaches that use similar visualization techniques or visual encodings as we do. For example, Henry et al. proposed NodeTrix [21], a hybrid visualization technique that enables users to select parts of the networks and to seamlessly switch between a matrix-based and a node-link visualization in a similar manner as in our approach. Another closely related system is g-Miner [13], which also provides an adjacency matrix and a node-link diagram to represent network data. Both representations use similar encodings, such as pie charts to depict different attribute values in the node-link diagram or a color mapping to represent relationships in the matrix-based visualization. However, g-Miner was developed with the intention to facilitate the visual analysis and interactive refinement of groups that are based on multivariate

data. With our approach, we focus on the comparative analysis of character networks in narrative texts and support an in-depth text analysis of the relationships between the characters.

There are also many approaches that particularly support the visual character network analysis in the context of the DH. Klein [28] proposed an approach that enables users to illuminate extracted named entities of historical texts about slavery using an arc diagram. Jänicke and Focht [22] offered an approach that uses a node-link visualization to enable the exploration of relations between musicians. Trilcke et al. [40] also uses node-link visualization to investigate the dramatic in texts. John et al. presented the system ViTA [24] that facilitates the character analysis in narrative texts. It supports different visual abstractions, such as word clouds or character networks, which are linked to the text and thus enable users to inspect the corresponding text passages in detail. Another approach that focuses on the exploration and analysis of named entities and terms co-occurring with them is POSvis [44]. It uses NLP methods to extract named entities and co-occurring part-of-speech (POS) tags. Multiple coordinated views facilitate users in analyzing entities in detail.

Very closely related to our approach is the work of Oelke et al. [33], which uses the distant reading technique “literature fingerprints” [26] to show the evolution of extracted characters and their relationships. It uses an adjacency matrix where rows and columns represent the characters, and the cells contain fingerprints of the relationships between the respective characters. However, this approach focuses on the comparison of two characters with each other, and it is difficult to understand the relationships between several characters in the adjacency matrix.

3 APPROACH

Our web-based approach provides valuable support for exploring and comparing character constellations within different selected text ranges of a text document. In the following, we discuss the requirements as they can be inferred from the basic problem, we explain the data preprocessing steps, and lastly, we present the different visual views and interaction possibilities of our approach.

3.1 Requirement Analysis

Our approach was developed in a formative process and in close collaboration with our project partners. Currently, the humanities scholars involved in the project visually analyze their annotated named entities with the open source tool Gephi [5]. It provides a wide range of methods that support the analysis of graph-based structures. However, Gephi has some shortcomings regarding the visual comparison of character networks and is therefore limited concerning the needs of our humanities scholars (see Section 5).

Based on these shortcomings, we defined in close cooperation with the humanities scholars during several internal workshops the following goals: **(G1)** The visual approach should enable users to select different ranges of a text document and it should visually represent the character networks described within them. **(G2)** It should support a comparison mode that enables users to compare selected character networks with each other to highlight the differences and commonalities of them. **(G3)** The character network visualization should be equipped with interactive features to facilitate the focusing and filtering of different parts of the network, for example, to highlight characters that only occur in one of the selected ranges. **(G4)** The approach should provide suitable visual comparison representations for smaller and larger character networks. **(G5)** The character network representations should be linked to a text view, which enables a close reading approach. **(G6)** It should be possible to analyze the relationships between characters in more detail, for example, to show a textual summarization of their contexts. **(G7)** The visual approach should also provide a similarity overview of all

selected character networks that serve as a starting point for a more in-depth analysis.

3.2 Data Preprocessing

On the one hand, the approach supports the analysis of text documents that were annotated by our project partners. The involved humanities scholars have developed an annotation scheme and annotated the text documents following an inter-agreement workflow. They annotated all characters and locations as well as further terms, such as verbs and adjectives that describe the relationships between those entities. On the other hand, in case that no annotations are available in a text, we process it along an NLP pipeline to automatically extract the information needed. For this purpose, we use the StanfordCoreNLP toolkit [31] to apply tokenization, sentence splitting, named entity recognition (NER), and (POS) tagging. With the aid of NER, we can automatically extract named entities (we only focus on characters) and with POS tagging we can extract the adjectives and verbs of the text. Afterward, we use a sentiment lexicon [30] to assign a negative, neutral, or positive meaning to each of the annotated or automatically extracted verbs and adjectives.

As a next step, we define when annotated characters are related. Based on experiments and discussions with our humanities scholars, we define different text ranges for a relationship. For example, for *Parzival*, we define that two characters or a character with a verb or adjective co-occur when they are mentioned within one paragraph and for the computationally analyzed text documents, we use a text range of one sentence. Subsequently, we use the javascript toolkit D3 [8] to calculate a force-directed layout for the node-link representation of the character networks over the whole text document.

3.3 Visual Approach

The main workspace of our approach is depicted in Figure 1. On the left side, there is a menu (see Figure 1 A) where users can load a new file and switch between a matrix-based and a node-link visualization or choose a grid view that provides an overview of selected graphs. We offer four different visual encodings for the node-link diagram and three for the matrix-based visualization in the comparison mode. Furthermore, users can activate a sentiment mode or apply filter techniques that support the exploration of the comparison of the character networks.

In the center of the workspace, either the matrix or the node-link view is shown (see Figure 1 B). At the top, both representations are complemented with document bands that represent the loaded documents and enable users to select a certain range of them **(G1)**. The respective selection brushes are represented in the color of the generated graph and the gray lines represent, for example, parts or chapters of the document. By hovering over the gray lines, we show the name of the segment in a tooltip. In case that the sentiment mode is activated, we show an additional sentiment band below the document bands where green bars represent positive and red bars negative verbs or adjectives that co-occur with a character or in the context of a selected pair **(G6)** (see Figure 1 B). This way, we provide semantic information about the characters, the relationships between them, and their evolution during the plot. For overlaps of bars, we use a color transition between the two colors to depict the wavering sentiment results. Furthermore, users can copy the character network from the main view and add it to the graph desktop (see below) via the menu on the left side. We encode each added character network with a unique color, using a qualitative color scheme of ColorBrewer [11]. From the graph desktop, it is possible to compare the selected graphs with the main graph **(G2)**. In the following, we introduce and explain the three different visual representations and the comparative possibilities in detail.

Node-Link diagram: The node-link representation is a widely used technique to visualize smaller graphs in an easily understand-

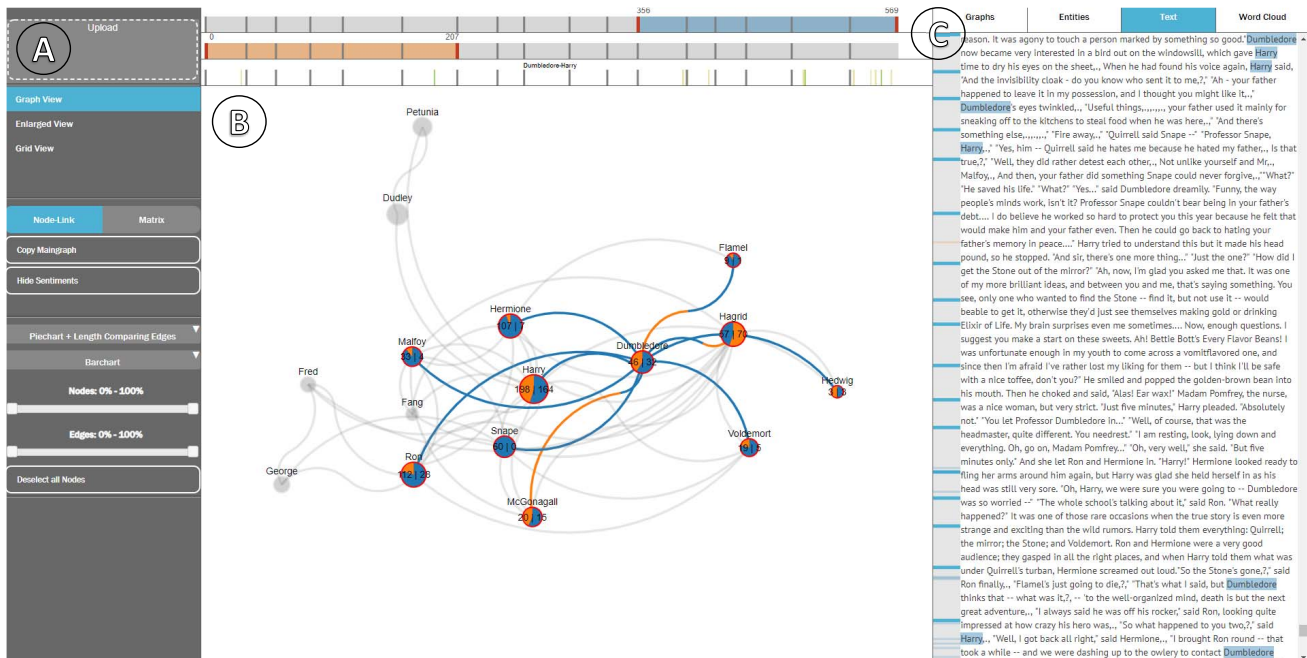


Figure 1: The main workspace comprises (A) a menu that enables users to switch between the different visualizations or to filter the character network, (B) the main view where users can inspect the character networks with the aid of matrix-based or node-link visualizations, (C) the tab view where users can switch between a graph desktop, an entity list, a text view, and a word cloud view.

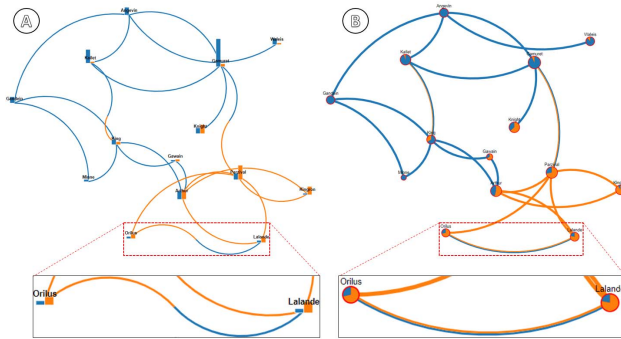


Figure 2: Four different node-link diagram encodings: (A) juxtaposed bar charts with length comparing edges and (B) pie charts with width comparing edges.

able way [17] (G4). It consists of vertices (characters) and edges (relationships) to represent the graph structure. Users can use either a bar or circle to represent characters. The size of the circles and the length of the bars are scaled proportionally to the characters' occurrences. In the node-link comparison mode, we offer bar or pie charts to illustrate the distribution of the characters' occurrences in the selected parts of the document as represented in Figure 2 (G2). We map the number of the characters' occurrences to the length of each bar (see Figure 2 A) or to the pie charts slices as depicted in Figure 2 B.

To represent the relationships between the characters, we use curved lines, and in the comparison mode, users can choose between two different visual encodings. On the one hand, we offer length comparing edges (see Figure 2 A) and on the other, we support parallel lines as width comparing edges as depicted in Figure 2 B (G2). We also scale the width of the parallel lines proportionally to the

characters' co-occurrences in the different character networks. Regarding the length comparing edges, we split the edges into two parts with a vault to indicate the transition and proportionally scale the length of each part to the relative co-occurrences of the characters.

With the aid of these visual encodings, users can get an overview of the characters and their relationships as well as the possibility to inspect the differences and commonalities of the character network (G2). Furthermore, it is possible to double-click on an entity or relation in order to analyze the corresponding text passages in the text view (G5), the related verbs and adjectives in the word cloud view (G6), or the sentiment results in the sentiment band (G6). Although we use curved lines to avoid ambiguous lines, the legibility of the character network can deteriorate significantly with many vertices and edges. More precisely, overlaps between vertices or edges can easily arise, which leads to occlusion problems (visual clutter) [4]. Thus, the visual exploration of the graph for users can get confusing. To alleviate the occlusion problems, we provide filter techniques to support the exploration of the character networks (G3). By hovering over a character, all characters and relations not directly connected to it are grayed out as depicted in Figure 1 B. Users can gray out particular characters and their corresponding relations by clicking on them. Additionally, users can set the percentage to filter characters or relationships that only occur in one of the selected text ranges or that frequently occur in both via the sliders on the left side menu (see Figure 1 A). For example, users can set the percentage to 100% to filter characters or relationships that only occur in the different selected ranges of the text document. Thus, users can quickly identify and inspect constellations of characters within the selected parts. However, these filter techniques do only support users up to a certain extent. Therefore, we also provide a matrix-based visualization that eliminates visual clutter and thus facilitates the possibility to analyze larger character networks [17].

Adjacency matrix: In contrast to the node-link view, the adjacency matrix provides a more scalable approach to analyze large networks (G4). The rows and columns of the adjacency matrix rep-



Figure 3: Three different visual encodings for the comparison mode of the adjacency matrix: (A) juxtaposed color-coded bar charts, (B) split crosswise cells, and (C) color transition.

resent the characters and the cells their relationships. If the user is not in the comparison mode, we use a color gradient from white to the respective color of the graph to represent the number of co-occurrences between two characters as depicted in Figure 6 B. A rather bright cell represents only a few or no co-occurrences whereas a dark one represents many. In the comparison mode, we offer three visual encodings to highlight the differences and commonalities of selected character networks as depicted in Figure 3 (G2). We provide juxtaposed color-coded bar charts (see Figure 3 A) where the length of each bar is scaled proportionally to the absolute occurrences of the characters. Additionally, we offer the visual encoding split crosswise, where we use diagonal divisions of the cells and again a color gradient from white to the respective color of the separate network as depicted in Figure 3 B. As a third option, we support a color transition of both selected character network colors as shown in Figure 3 C. As with the node-link visualization, we offer interaction possibilities. Users can hover over a cell to show the number of co-occurrences in a tooltip, and by clicking on an entity label or cell, we depict the occurrences or co-occurrences of the characters in the text view (G5) as well the corresponding verbs and adjectives in the word cloud view (G6). In addition, users can sort the matrix by name or frequency of the characters. We also offer the possibility to filter characters and their relationships that, for example, are only mentioned in one of the selected parts via the sliders on the left side menu (G2).

Furthermore, we allow users to examine the annotated or extracted verbs and adjectives, which describe the characters' and their relationships' traits, directly in the matrix-based visualization (G6). To facilitate this, we provide an integrated focus+context (fisheye) technique, which is similar to the Table Lens approach [35]. The fisheye approach uses a spatial distortion in order to – within a single view – focus upon particular areas without losing their context. Thus, focused areas show a high level of detail, whereas non-focal regions show a low level of detail but are still visible. In our focus+context matrix approach, users can choose between three different zoom levels, using the mouse wheel. We use the available space in the focused regions to represent the verbs and adjectives as word clouds in the respective cells as depicted in Figure 6 A. The terms of the word clouds are sorted by their frequency, and in the comparison mode, we use bars below the words in the colors of the selected parts to depict the distribution. In case that the sentiment mode is activated, we depict negative terms in red, neutral terms in black, and positive terms in green. That way, we enable users to freely navigate and explore the different word clouds pleasantly without scrollbars within a single view. However, if the space of the cells becomes too small to display a word cloud, we use one of the three

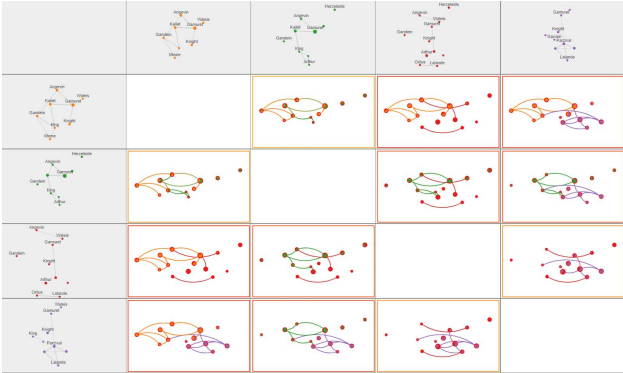


Figure 4: The grid view provides an overview of all selected character networks as juxtaposed small multiples in an adjacency matrix. The frames' colors represent the similarity between the respective character networks.

mentioned visual encodings as depicted in Figure 6 A. In case that the cells also get too small for the juxtaposed bar charts or split crosswise visual encodings, we depict the color transition in these cells to give at least a small hint to the user.

Grid view: The grid view provides an overview of all selected character networks as juxtaposed small multiples (see Figure 4). The rows and columns of the grid view show all selected graphs, and the cells represent the respective comparison graphs similar to the approaches of Blascheck et al. [7] and Burch et al. [12]. To facilitate the comparison of the graphs, we use a simple measure to calculate the similarity between them (G7). The similarity measure compares the number of characters' occurrences and co-occurrences between two selected character networks. Subsequently, we map the similarity values to values on a color gradient from red to green. To show the similarity in the grid view, we use a frame in the cells, where green colors indicate similar and red colors very different graphs as depicted in Figure 4. That way, the grid view provides a comparison overview of all selected graphs and can serve as a starting point for further analysis. For example, users can find an interesting comparison graph and further analyze the graph in the main view by double-clicking on the cell.

Tab view: On the right side of the workspace (see Figure 1 C), users can switch between four tabs: a graph desktop, an entity list, a text view, and a word cloud view. The graph desktop contains small multiples of graphs corresponding to the text ranges selected by the

users in the form of a matrix or node-link representation, depending on the selected mode as depicted in Figure 6 B. Users can also pan, zoom, and rearrange the small multiples. However, the labels of the characters can quickly become unreadable because of the occlusion problem. Therefore, we check for overlaps during zooming, and in case of an overlap, we only show the name of the entity that appears more frequently. With drag and drop, users can focus the small multiple in the main view. By clicking on the arrow (see top right), users can compare a small multiple with the character network in the main view.

The entity list provides an overview of the characters sorted by their frequencies of occurrence and enables users to select and deselect them from the visualizations dynamically as depicted in Figure 5 (G3). Additionally, we show for each added character network a pixel-based fingerprint visualization to represent the distribution of the characters. Here, each cell represents a particular part or chapter of the text document, and the color saturation indicates the number of occurrences. Light colors represent only a few occurrences, whereas saturated, dark colors represent many occurrences of the entity. In the case that users have only selected a sub-range of the text document, we highlight the respective cells with a red frame to indicate the selected range.

The text view is linked with the adjacency matrix and the node-link visualization and allows users to work directly with the text (G5). We use a sidebar to highlight the occurrences (or co-occurrences) of selected (pairs of) characters next to the text view as depicted in Figure 1 C. The blue bars represent the parts of the text document, and (co-)occurrences of characters are highlighted in the color of the respective selected character network. By clicking on an occurrence in the sidebar, users can navigate to the corresponding text passage. Additionally, we highlight the characters in the text as well. That way, users get an overview of the distribution of the occurrences or co-occurrences of the selected characters and a possibility to easily access them.

The word cloud view is also linked with both main visualizations and represents extracted or annotated verbs and adjectives that co-occur with selected characters in the chosen text ranges (G6). As with the focus+context matrix view, we use bars to provide an overview of the distribution of the terms in the comparison mode, we color the terms according to their sentiment assignment (in the sentiment mode), and we offer the possibility to further analyze the words in the text view by clicking on them (G5).

4 USAGE SCENARIOS

In the following, we present two usage scenarios. The first one is based on a fictitious analysis of the modern novel *Harry Potter and the Philosopher's Stone* by J. K. Rowling. Afterward, we present an analysis of the Middle High German romance *Parzival* by W. von Eschenbach which is based on insights gained in joint workshops with our project partners.

4.1 Analysis of *Harry Potter and the Philosopher's Stone*

In this usage scenario, we want to study the commonalities and differences of the character constellations between the first and second half of the novel *Harry Potter and the Philosopher's Stone*. After having loaded the text document into our approach and after it was preprocessed successfully, we can see the respective node-link diagram in the main view. In the first step, we roughly analyze the character network. We can quickly identify the main characters of the novel, such as *Harry*, *Ron*, or *Hermione*. To compare the first with the second half of the novel, we select the first six chapters in the document band and copy the selected graph to the graph desktop (G1). Afterward, we select the last six chapters in the main view and compare the resulting graph with the small graph on the graph desktop.

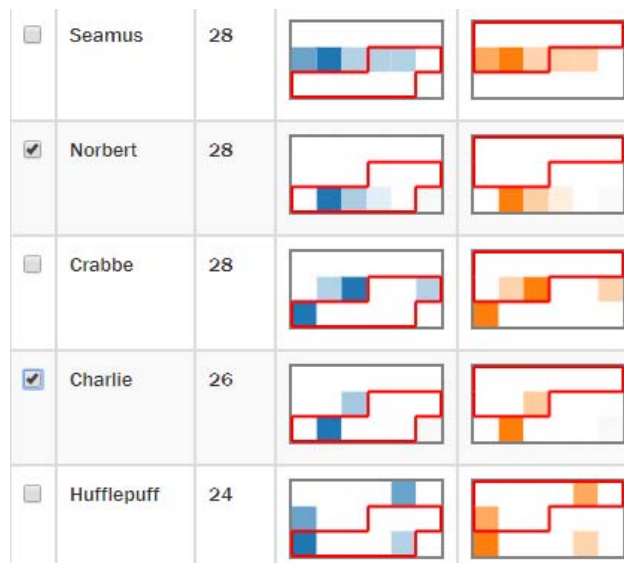


Figure 5: The entity list enables users to select and deselect characters from the graph, shows the number of characters' occurrences, and fingerprint visualizations that represents the distribution of the entities in the different parts of the text document.

In the next step, we further explore the character network. We may have immediately identified many characters that we do not want to analyze further. Therefore, we switch to the tab "entity list" to filter them. Here, while scanning the list, we detect that the NER method assigned wrong tags to a few characters, for example, the organization *Hufflepuff* was marked as a person. Additionally, with the aid of the complemented fingerprint visualization (see Figure 5), we identify characters, such as *Seamus*, or *Crabbe*, that mainly appear in the middle of the book and hence are also irrelevant to our analysis; we deselect them (G3).

During the analysis, the entity *Fang* has aroused our interest. We are uncertain whether to show or hide it because we are unsure about its role in this text. To gain more information about *Fang*, we double-click on the respective node in the node-link diagram and switch the tab with the word cloud view in order to analyze the verbs and adjectives that co-occur with *Fang* (G6). By analyzing the word cloud view, we quickly get the impression that it must be an animal because of terms like "hairy", "tame", or "hungry". Afterward, we switch to the text view tab to inspect some text passages that contain *Fang* (G5). With the help of the sidebar and the possibility to directly access the corresponding text passages, we quickly find out that *Fang* is the dog of *Hagrid*, the gamekeeper, and we keep the selection.

After these first selections, we now want to visually compare both selected parts of the novel in the main view. We consider the node-link representation with the option "pie charts with length comparing edges" and quickly recognize a series of differences between the two parts (G2). With the help of the pie charts and the filter function to highlight characters that mainly appear in one of the two parts, we notice that in the first half of the book mostly characters from the "muggle world" occur and in the second half mostly characters from the wizard world. Furthermore, using the "length comparing edges", we can conclude that the two worlds are primarily linked by the characters of *Harry*, *Hagrid*, and *Hedwig*, the pet of *Harry*.

As a next step, we want to analyze the relationships of *Dumbledore* with the other characters. To this end, we hover over *Dumbledore* to gray out all characters and relations not directly connected to him as depicted in Figure 1 B. Afterward, we select only the

relevant characters and activate the sentiment mode to inspect the relationships in more detail. After selecting and scanning the different relationships with the aid of the sentiment band, we find out that most of them are very positive. For example, the relationship between *Harry* and *Dumbledore* is mainly positive (see Figure 1 B). To further analyze the semantic information, we can click on occurrences to jump directly to the corresponding text passages in the text view as depicted in Figure 1 C.

4.2 Analysis of *Parzival*

In the second usage scenario, we consider an inquiry that was conducted by our project partners. They used our approach to analyze the Middle High German text *Parzival* by W. von Eschenbach, which they had annotated manually. In order to ensure a better understanding of this inquiry, we reproduce it here on the basis of an English translation of the text by J. Weston, where we extracted the respective entities by means of NLP. Therefore, the use case's results differ slightly from the original inquiry's; for example, the characters' occurrences can vary slightly, or a few characters were detected wrongly. However, the overall outcome of our project partners' original analysis is reflected in the results of the analysis laid out in the following. The participating humanities scholars have a strong background knowledge about the text, whereas the computational linguists have only little previous knowledge. In the following, they want to confirm their knowledge using our approach.

After we loaded a text document, which contains the first nine books of *Parzival*, into our system, they start their analysis. First of all, they explore the characters of the whole book in the node-link diagram and retrieve all of the essential characters. Afterward, they want to compare the characters' co-occurrences within the first three books with their co-occurrences within the four following books. Therefore, they create a second graph and select the different parts in the respective document bands (G1). To compare the characters' co-occurrences, they switch to matrix-based visualization as depicted in Figure 3 (G2). After inspecting some co-occurrences (with the help of the tooltips), the analysis results confirm their expectations.

As a next analysis step, the humanities scholars want to inspect the evolution of the characters' networks over the course of the first four books in order to verify their hypothesis that the character networks are not similar. Therefore, they create four different character networks and add them to the graph desktop as depicted in Figure 6 B (G1). Subsequently, they select the node-link diagram visualization and switch to the grid view in order to compare the different character networks with each other as depicted in Figure 4. With the aid of the similarity function, they can quickly confirm their hypothesis (G7).

As the next step, they want to analyze the different comparison character networks in more detail. Therefore, they focus on the comparison graphs, one after the other, in the main view by double-clicking on the respective small multiple (G2). They find out (which confirms their assumption) that the different character networks are only connected via a few entities, such as *Parzival* or *Arthur* as depicted in Figure 2.

In order to get more information about the co-occurrences of the characters, they switch back to the matrix-based visualization and activate the focus+context mode using the mouse wheel as depicted in Figure 6 A. While they freely navigate and explore the different word clouds, they come across terms which they did not expect (G6). Those terms could now serve as a new starting point to analyze them more deeply.

These usage scenarios show that our approach supports the comparative visual analysis of characters of two selected passages in a text document.

5 EXPERT FEEDBACK

This section provides feedback and insights from two-day workshops that are held jointly with our project partners from the humanities twice a year as well as project meetings that are held every two weeks. Based on a separate interview with two project collaborators, we present additional insights into the benefits and drawbacks of our approach compared to Gephi. The workshops and meetings serve as platforms to discuss open questions, for example, about the annotation guidelines or the workflow, and they offer the possibility to present approaches that are currently being developed in the context of the project. Thus, we could gain feedback about features we implemented and could tailor them further according to the users' needs to support specific analysis tasks.

General Feedback: Overall, during the project meetings, there was always a great interest in this approach since our partners have to deal with this kind of analysis tasks on a daily basis. The involved humanities scholars perceived the approach as understandable and easy to use without a long training period. Initially, the node-link visualization was of greater importance for them, because they were already familiar with this kind of representation from Gephi. However, over the course of the development and after some discussions and explanations from our side, they became more and more open to the matrix representation. Eventually, they began to use the adjacency matrix and to switch between the views more often, e.g., for analysis tasks involving many characters, which would lead to visual clutter in the node-link visualization.

Regarding the usefulness of the different visual encodings to represent the characters' occurrences and co-occurrences in the node-link comparison mode (G2), the experts were quite discordant. They could not distinguish any advantages of the bar over the pie charts or vice versa, because both representations provided a quick overview and the experts would use the tooltips to inspect the exact value. However, one humanities scholar found the pie charts more visually appealing. Regarding the co-occurrences, the experts preferred to work with the "length comparing edges". In this representation, the edges were easier to track and to compare with each other because it caused less visual clutter. Additionally, they were perceived as more aesthetic than with the "width comparing edges". Furthermore, one of the humanities scholars mentioned that the different filter techniques (G3) and highlights to depict the differences and commonalities of the character networks (G2) would help her to explore and analyze the node-link diagram. Especially the filter operation to focus characters that only occur or co-occur in one of the selected parts would facilitate her work (G3).

The graph desktop list was considered to be very helpful in their workflow because caching character networks and retrieving them later for further analysis is an essential working step for them. Additionally, the humanities scholars emphasized the advantages of the combination of the graph desktop list and the grid view, because the stored character networks can be directly compared with each other. With the aid of the similarity method, it would be possible to detect new interesting character networks for a more in-depth comparative analysis (G7).

The entity list with the complemented fingerprint visualization, however, was only considered helpful for the selection and deselection of characters (G3). The humanities scholars commented that the fingerprint visualization would provide a good overview, but that it would be very challenging to compare entities, especially those who are situated far away from each other in the list. They would have liked to have a comparison mode for two selected characters or at least the possibility to reorder the entities, e.g., with a drag and drop interaction.

The humanities scholars liked the opportunity to access the text view directly from the interrelated visualizations (G5). In particular, they liked the sidebar which highlights the occurrences or

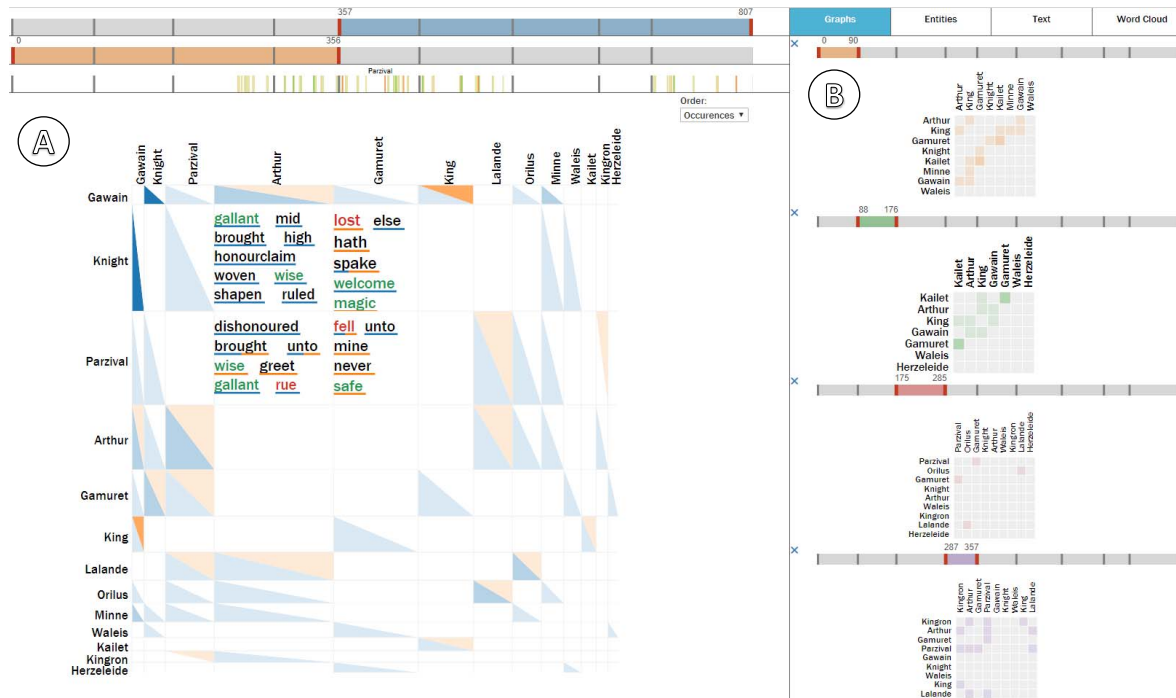


Figure 6: (a) Our focus+context approach in the comparison mode with two different selected parts of the romance Parzival. (b) The graph desktop containing four user-selected graphs.

co-occurrences and enables users to jump to the corresponding text passages. However, they mentioned that a possibility to analyze selected text passages next to each without scrolling would be very beneficial.

The possibility to inspect the verbs and adjectives in the word cloud view was also rated as advantageous (G6). That way, the humanities scholars could analyze terms that co-occurred with the selected and thus they could get a fast overview and impression about the entities. However, they would rather have liked an integrated word cloud within the node-link view, similar to the focus+context matrix view.

The integrated focus+context technique in the adjacency matrix is one of the newest features of our approach. The preliminary feedback we received was very positive and indicates its usefulness for understanding (changing) contexts of relations (G6).

Comparison with General Workflow: Currently, the humanities scholars use Gephi to visually analyze and compare the annotated character networks. To gain insights about how our approach can facilitate and improve their general workflow, we decided to conduct an additional expert interview with two project partners that have worked a lot with Gephi for the last two years.

Both participants mentioned that the possibility to select freely between different ranges of a text document (G1) and to visually compare the character network within them in one compact view is an advantage of our approach. In this context, they also found that the visual encodings would facilitate the identification of the differences and commonalities of the selected ranges of the text document in comparison to Gephi (G2). Thus, they no longer need to scan character networks next to each other in order to compare the different occurrences of the characters and their relationships. Especially in novels with several storylines, such as in the *Parzival* romance, it is interesting to see which characters act in both of the selected parts and which are assigned to a specific storyline.

One participant remarked that Gephi offers more network metrics such as centrality or density and different layouts for network representation. However, such features could be easily integrated into our approach.

They liked the opportunity provided by our approach to switch seamlessly between a node-link diagram and a matrix-based visualization, which is not supported by Gephi (G4). Another necessary missing feature for the humanities scholars is that Gephi does not support the opportunity to access the text or a word cloud view directly from the visualizations (G5). This would be especially helpful in order to explore the contexts of relationships that they did not expect.

Both participants, however, mentioned that our approach would not support an animation that shows the evolution of the characters and their relationships in the form of a movie. To realize such a feature, we could allow users to select a text range for the displacements, as well as the duration for the individual steps.

Furthermore, the participants liked the grid view because it is interesting to see the similarity between selected networks which in turn is not supported by Gephi (G7). Overall, concerning their analysis, both participants saw clear advantages for our approach as compared to Gephi.

6 DISCUSSION

After the discussion of the experts' feedback, we now want to illuminate the scalability of our approach and discuss our design choices as well as the limitations for the visual comparison of character networks with our approach.

Scalability: Our approach is scalable concerning several aspects. Firstly, it can deal with arbitrarily long text documents that contain many entities. The node-link diagram, however, can quickly get cluttered if many characters are selected. To tackle this problem, we provide different filter techniques and the possibility to switch

to an adjacency matrix view, which is scalable for larger graphs. The other views also have no scaling problems in this context and therefore they can deal with arbitrarily long text documents and much additional information.

Secondly, our approach can be extended to facilitate the comparison of several text ranges with each other. Currently, the adjacency matrix and the node-link diagram only support the comparison of two selected parts of the text document. However, the pie and bar charts, which depict the occurrences and co-occurrences, can represent more parts without fundamental changes. A disadvantage would be that the bars and the slices of the pie chart would become smaller and smaller and therefore more difficult to discern. Both edge representations in the node-link diagram can also be adapted to represent more different selected parts without fundamental changes. However, the visual encodings “split crosswise” and “color transition” in the matrix view do not scale as well. For the “split crosswise” encoding, it would still be conceivable to divide the cells into four sections; however, this could lead to patterns that interfere with the cell boundaries of the matrix [4]. Concerning the “color transitions”, the more different colors contribute to a mixture, the more difficult it gets to defer each color’s relative contribution to the mixture – it may be even too difficult for more than two colors. Since each part is color-coded, we would quickly run out distinguishable colors. An alternative would be to use glyphs instead of colors. However, these would require more space and a learning phase on the user’s side.

Instead of comparing different ranges, we could also compare different versions of a text document. The two tasks are similar and all of the discussed points above also apply for this new task, but now we would have to adjust the length of the document bands and the fingerprint visualization in the entity list view. A more complex extension would be the support of a comparative text analysis because this is essential for this task. One possibility would be to show the different text versions in juxtaposed text views and enable users to change the order of the documents by drag and drop. However, this would require much space and thus the analysis would be limited to a small number of different texts.

Design Choices and Limitations: During the development of our approach, many choices had to be made to fulfill the requirements of our project partners (see Section 3.1). In the following, we discuss the design choices and the limitations of the used visualizations for the character network comparison.

We offer a node-link diagram for the visual comparison because it is a widely used technique to visualize networks, and the humanities scholars are familiar with it. Concerning the layout of the node-link diagram, we decided to calculate a force-directed layout over the whole text document (instead of the selected segment) since fixed positions of the characters simplify the comparison of character constellations in different selected text ranges. To represent the different occurrences of the characters in the selected text ranges, we offer either pie charts or bar charts. We decided to use pie charts as they fit the conventional node-link representation and thus provide a good transition between nodes and edges. However, we also decided to offer bar charts because they support a better comparison of the different occurrences of a character as well as of the absolute character occurrences. Concerning the visual encodings for the relations between characters, we provide “length comparing edges” and “width comparing edges” as depicted in Figure 2. We decided to offer “width comparing edges” because they are easy to understand. However, the “width comparing edges” have the disadvantage that the parallel lines can consume more space and thus the network can be confusing. Therefore, we also support “length comparing edges” because they cause less visual clutter. However, depending on the vertices’ positions, the impression could arise that longer edges represent more co-occurrences than shorter edges (which is also due to the force-based layout). We are aware that we have to

communicate this limitation in detail to the humanities scholars to avoid misinterpretations and we leave the choice to the user. Another design decision was not to integrate word clouds within the node-link view, similar to the focus+context matrix view, because this would lead to additional visual clutter. Instead, we decided to show a word cloud in a separate view with the limitation that the users have to look back and forth.

In addition to the node-link diagram, we decided to provide an adjacency matrix because it provides a more scalable approach to analyze larger character networks. As with the node-link visualization, we support different visual encodings to represent the different occurrences of the character relationships in the selected text ranges, which have their benefits, but also limitations. We offer “juxtaposed color-coded bar charts” that support a good comparison of cells in the respective vicinity. However, with many cells, they can become very small and thus difficult to compare. Therefore, we decided also to support the “split crosswise” and “color transition” visual encodings which are based on a color mapping and provide a better comparison of smaller cells. Additionally, both can be integrated into the focus+context approach without any disadvantages. To investigate the relationships in more detail, we decided to integrate a focus+context mode which represents the word clouds of characters in the respective cells. This way, users do not need to look back and forth. However, it must be noted that due to the distortion, the bars of the bar chart encoding are no longer absolutely comparable and thus are only partially useful in the focus+context mode: While the relative comparison within the column or row of the focused cell is still possible, bars that are farther away are sized on a different scale. This is also a limitation that we have communicated to the humanities scholars to avoid confusions.

7 CONCLUSION AND FUTURE WORK

In this work, we present a visual approach for the comparative analysis of named entity networks in narrative texts. Our approach allows users to select two different ranges of a text document and then to compare the character networks represented within them. This is accomplished by the means of a comparative graph visualization, explicitly encoded either as a node-link diagram or as an adjacency matrix. Additionally, we support interactive and visual methods that enable users to filter the respective graphs, to focus the corresponding text passages, or to analyze the characters and their relationships in more detail with the aid of a focus+context approach, sentiment analysis, and word cloud visualization. Two usage scenarios demonstrate the usefulness of our approach, and finally, we present and discuss expert feedback that arose during our project meetings.

During the close cooperation with our humanities scholars, many ideas for future improvements were discussed. For example, we want to extend our approach such that we can support the analysis of different relationship definitions. Currently, the humanities scholars annotate context-based text segments to compare the resulting character networks with character networks that are based on text structure, such as paragraphs or strophes. Another interesting direction we would like to pursue is to support the analysis of multiple text ranges or different versions of a text document. Furthermore, we intend to make our approach available online, such that external analysts can analyze their texts.

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